Tahsin Touheed Ivan

+880 1302-535848 | ivan19tauhid@gmail.com | LinkedIn | GitHub

Education

University of Dhaka — Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering (4th year) — *Sept 2022 – Dec 2026 (expected)*
CGPA: 3.73 / 4.00

Notre Dame College — Dhaka, Bangladesh
Higher Secondary Certificate (Science) — *Aug 2019 – July 2021*
GPA: 5.00 / 5.00

Ideal School & College — Dhaka, Bangladesh
Secondary School Certificate (Science) — *Jan 2009 – July 2019*
GPA: 5.00 / 5.00

Research

Materials Science Research Laboratory (MSRL DU) — Jan 2026 – Present
Undergraduate Researcher | Supervisor: Prof. Dr. Imtiaz Ahmed
Interests: ML-guided electrode design and materials discovery for energy storage, with broader curiosity in nanomaterials characterization and micro/nano-engineered devices.

Current Work — *Machine-Learning & DFT-Guided Electrode Design for Oxide-based Supercapacitors: From Data Mining to Experimental Validation*

- Building a Computation-guided Closed-Loop Pipeline for perovskite-oxide (ABO_3 and $A_2BB'O_6$) supercapacitor electrodes. In the first phase, a literature-mined descriptor dataset — spanning chemistry, structure, synthesis, and testing condition features — is curated, feeding ML models (linear and tree based ensembles) that rank compositions by specific capacitance (Csp) and stability, with SHAP/PDP explainability surfacing the dominant drivers; the resulting shortlist is then cross-checked against first-principles DFT (band gap, formation energy, and related stability descriptors).
- Experimental validation is planned as the second phase: ML-identified high-impact features will guide targeted material synthesis, followed by structural characterization (BET, XRD, Raman, XPS) and electrochemical evaluation (CV, GCD, EIS) — closing the loop between data-driven prediction and physical material behavior.

Leadership

IEEE SIGHT Student Branch, DU — Feb 2025 – Present

Chair — *April 2026 – Present*
- Led the team that secured the IEEE HTB Tech4Good 2026 grant (USD 4,739) for *SmartDrip*, an IoT-enabled drip irrigation system for urban rooftop plantation; currently leading project execution end-to-end.
- Supervising collaborations with IEEE chapters on technical workshops and seminars to expand technical and community impact.

Program Coordinator — *Feb 2025 – March 2026*
- Coordinated and organized the *Machine Learning Fundamentals* workshop for undergraduate students — managed logistics, content planning, and participant engagement.
- Led a hands-on session in the *IEEE Day Outreach Programme on Microcontrollers and Their Utilization* — jointly organized by IEEE SB University of Dhaka, IEEE EDS SBC DU, and IEEE WIE AG SB DU at Nawab Faizunnesa Govt Girls High School, Cumilla — for 50+ high school students.

IEEE Photonics Society Student Branch, DU — Feb 2025 – March 2026
Event Coordinator
- Organized technical workshops, seminars, and community outreach programs.

Optics and Photonics Club Jan 2024 – April 2025

Vice President

- Worked on projects, seminars, and events hand-in-hand with OPTICA.
- Delivered talks and project showcases (LFR, Holography, etc.) at the outreach program *Envisioning Harmony Between Human and Nature*, St. Joseph Higher Secondary School, Dhaka — in collaboration with the Optica Bangladesh Student Chapter, University of Dhaka, and Scintilla Science Club.

Projects

SmartDrip: IoT-Enabled Drip Irrigation System for Urban Rooftop Plantation | *Embedded Systems* 2026 – Ongoing

Awarded: ***IEEE HTB Tech4Good 2026 Grant (USD 4,739)***

- Leading an IEEE SIGHT SB DU team developing an IoT-enabled drip irrigation system aimed at making urban rooftop plantation water-efficient and low-maintenance for Old Dhaka residential area — addressing freshwater stress and food-resilience challenges in dense urban areas.
- Proposal selected for funding under the IEEE Humanitarian Technologies Board's Tech4Good 2026 call; currently directing system design, sensor–actuator integration, and field-deployment planning.

LC-Based Capacitive MEMS Pressure Microsensor | *Python, scikit-learn* 2026

- Built an end-to-end Python simulation of a passive, wirelessly-interrogated capacitive MEMS pressure sensor — modeling the full signal chain from membrane deflection to variable-gap capacitance, LC resonance, and inductive readout.
- Trained a neural-network surrogate on 171 valid geometries sampled via Latin-Hypercube, then applied inverse design to optimize the sensor's geometry for target performance ($\sim$600$\times$ faster than the full simulator, with every candidate verified on the true simulator).

Project: Drone (Funded by FabLab DU) | *Python, Arduino, ESP32* 2025

Supervised by: ***Dr. Md. Shafiul Alam***

- Co-developed FabLab DU's first departmental drone under $\sim$USD 66 — integrated an F450 Quad-X frame, A2212 1000KV BLDC motors, 30A ESCs, MPU6050 IMU, and FlySky FS-i6 RC system on an ESP32 DevKit V1 running ESP-FC/Betaflight firmware; achieved 2.8:1 TWR, $\pm 5^\circ$ hover stability, $\sim$15 m altitude, and $\sim$13 min flight time. Architected for extensibility with GPS, barometric altimeter (BMP280), FPV camera, and DSHOT ESC upgrade — without a flight-controller hardware change.
- Gained end-to-end embedded flight-systems experience: configured ESP32–ESC PWM communication (1000–2000 μs) and PPM/SBUS RC integration; implemented and hand-tuned cascaded PID control (rate + angle loops) across roll, pitch, and yaw; diagnosed and resolved ESC desync, IMU yaw drift, and vibration instability; explored ESP32's dual-core architecture and onboard Wi-Fi as a basis for telemetry and autonomous navigation — validating commodity microcontrollers as low-cost alternatives to STM32-based commercial flight controllers.

ESP32 Retro Gaming Console | *ESP32* 2026

- Built a handheld retro gaming console prototype (ESP32, 0.96" SSD1306 OLED, analog joystick, passive buzzer) running three original arcade games (Car, Snake, Pong) with a scrollable menu, idle sleep animation, and per-game high scores saved to flash.
- Focused on ESP32 real-time firmware design — millis()-based finite-state machines, ADC noise filtering with hysteresis, and debounce logic — alongside peripheral integration: I^2C display rendering with U8g2, LEDC PWM audio, and NVS for non-volatile storage across deep-sleep wake cycles.

Pocket Reader — Handheld E-Book Reader | *ESP32* 2026

- Built a self-contained handheld e-book reader prototype (ESP32, 0.96" SSD1306 OLED, analog joystick) serving a mobile-friendly web portal directly from the microcontroller for book uploads and Wi-Fi management — no cloud, no external server. Designed for extensibility toward an e-ink display and a distraction-free writing mode.
- Focused on ESP32 networking — Wi-Fi station and access-point modes, mDNS, and an on-device REST HTTP server — alongside NVS, LittleFS, and deep-sleep power cycles; reserved BLE hardware to preserve a future phone-sync feature without requiring shared Wi-Fi.

Laser Harp | *Arduino* 2023

- Sophomore academic project — played a two-octave span of musical notes using a laser-and-LED array.

Competencies

Programming: Python, C/C++, x86 Assembly, Verilog (RTL design), HTML, LaTeX
Software & Tools: PSPICE, Proteus, MATLAB, Simulink, AutoCAD, EasyEDA
Languages: English (IELTS 7.0; outdated now), Bangla (native)
Soft Skills: Public Speaking, Team Leadership, Mentoring, Event Management, Media Coordination & Direction

References

Dr. Imtiaz Ahmed
Professor, Dept. of EEE
Faculty of Engineering and Technology
University of Dhaka, Dhaka, Bangladesh
✉ imtiaz@du.ac.bd

Dr. Sharnali Islam
Associate Professor, Dept. of EEE
Faculty of Engineering and Technology
University of Dhaka, Dhaka, Bangladesh
✉ sharnali.eee@du.ac.bd

Additional references available upon request.